

blocks of matter, at the atomic and molecular levels. Think of taking each molecule one by one and arranging it to get what we want—something like Lego. This is exactly what Nanotechnology can do. Build from the basic building blocks.

Nanotechnology deals with particles of size in the nanometer range (10^{-9} metres). More precisely, it deals with particles of the size of 1 to 100 nanometers. These particles are manipulated and controlled to build materials to the standards we need.

This means that many of the micron level technologies being called Nanotechnology do not really fall into this group at all. When compared to a nano-sized particle, a particle of micron size is nearly 1000 times larger. That's almost like comparing a 1-metre diameter sphere to one with a diameter of 1 km. Or think of a nano particle as something solid, which is one-billionth of a meter in size. Ergo, think twice before reading or listening to someone who is talking about Nanotechnology in the micron range.

Nano tools

So how do we get down to the molecular level and tinker around with atoms? Researchers use tools such as Scanning Tunnelling Microscope (STM) and the Atomic Force Microscope (AFM) for this purpose. STMs and AFMs belong to the group called Scanning Probe Microscopes (SPM).

The STM has a very fine metal tip, which is used to get the images of surfaces to be scanned by applying voltages. When voltage is applied, electrons jump or 'tunnel' across the gap between the tip and the surface. This results in a small detectable current. The current will vary as the distance between the tip and the surface being scanned changes. Researchers get an accurate image of the particles (atoms and molecules) that make up the surface, their behaviour and properties this way. STM can also move individual atoms and molecules into desired positions.

There's a limitation when using STMs: the surface under study has to be a good conductor of electricity. AFM takes care of this since it involves actual contact between the tip and the surface for the imaging. Imaging of the surface is done by measuring the tiny force of contact which has to be kept constant while the tip moves up and down over the surface. This mode

of operation of the AFM is called the contact mode. In addition to this, AFMs also have the 'tapping' mode of operation where the tip is lowered to a particular point and then pulled up and lowered to the next point of the surface. AFMs are used to get the image of non-conducting surfaces such as biological tissues, plastics, glass, etc.

Now let's see what's possible with these tools at the nanometer level.

Nanocomputers

As of now, Nanotechnology is focusing on projects which can be implemented in bettering our lives. Pervasive computing is an area where a lot of Nanotechnology projects are currently active. For example, if you want to design a chip to fit into your fingertip for controlling a music system, then the solution lies with Nanotechnology.

While making a microprocessor, we handle big groups of semiconductor molecules and structure them into the form we need. This form of handling of matter produces severe limitations as to how small these circuits can be made. Present day lithographic technologies (used in the latest desktop processors) are at 0.13 microns. After the 0.13-micron limit, it is very difficult to etch the circuits precisely and effectively on the silicon substrate. This is where Nanotechnology steps in. Whether it is a matter of using DNA for computing or arranging semiconductor atoms to form logic circuits, Nanotechnology offers convenience hitherto unknown to bulk technology (most of today's technology can be termed as 'bulk technology' as it involves manipulation of atoms and molecules in bulk rather than one by one).

Computing giant IBM has come up with a new kind of memory called 'Millipede technology', which makes use of an array of AFM probes to make marks on a polymer surface for storing data. Each tip writes a 'bit' of 50 nanometer on the polymer which stores the data. Reading and erasing of data is also done by the same arrays. Today's best storage devices are capable of a memory capacity of 2 gigabits per square centimetre, whereas this technology has made possible densities up to 80 gigabits per square centimetre using a single AFM tip. This is about 40 times greater than the memory densities available now. These devices are not so far away either. One of the first steps in this direction was the Microdrive from IBM, which has 1 GB mem-

On the Nano Path

The term 'nanotechnology' was popularised by K. Eric Drexler in 1986 when he came out with a book called *Engines of Creation* in which he described what is possible with nanotechnology.

However, one of the first practical nanotechnological advances can be traced back to 1968, when Bell Laboratories developed molecular epitaxy, a technique to deposit single atomic layers on a surface. One of the most practical ways to effectively use nanotechnology happened in 1981 when Gerd Binnig deployed the scanning tunnelling microscope to image individual atoms. This was followed by the creation of Buckminsterfullerenes or buckyballs in 1985. Buckyballs measure about a nanometer in diameter and are used widely for nanotechnology applications.

Sumio Iijima of NEC discovered carbon nanotubes, yet another major nanotechnological advancement, in 1991 and in 1998 the first transistor was made from a nanotube by Cees Dekker. The fact that single molecules can act as molecular switches was demonstrated in 1998 according to research held at Rice University. The start of 2000 saw a great boost to nanotechnology in the form of the National Nanotechnology Initiative (NNI) in the US. NNI aims to provide more funding and greater public participation in the field of nanotechnology.

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